

Original scenarios and prior scenarios for evaluation

In the evaluation of the final mechanism consisting of point cloud initialization and supervised pretraining for the translation step and full fine-tuning for the correction step, various prior scenarios were used in connection with two original scenarios to be reconstructed. Some of these prior scenarios were generated directly with the RODOS system of the German Federal Office for Radiation Protection (BfS). Additional prior scenarios were specifically constructed to evaluate the effects and handling of certain errors in RODOS simulations by the method Cuboid Splatting. The original scenarios and the associated prior scenarios are presented below. This is followed by a discussion of the biggest uncertainty factors and sources of error in RODOS calculations. Finally, the targeted construction of prior scenarios for modeling certain types and intensities of errors in RODOS scenarios for the evaluation is presented.

For overview, here is a complete list of scenarios and prior scenarios:

Emsland

- Emsland (original scenario)

RODOS priors

- EuRim-Emsland prior

Constructed priors

- Emsland overestimation x2
- Emsland overestimation x10
- Emsland underestimation x0,5
- Emsland underestimation x0,1
- EuRim prior x0,8
- Emsland time-dependent source term error (x2, x0,5)
- Emsland time-dependent source term error (x10, x0,1)
- Emsland local weather turbulences
- Emsland systematic wind speed error
- Emsland area-wise small random noise
- Emsland area-wise medium random noise
- Emsland area-wise strong random noise
- Emsland perfect
- Brokdorf instead of Emsland (completely wrong)

Brokdorf:

- Brokdorf (original scenario)

RODOS priors

- Brokdorf_Prior1
- Brokdorf_Prior2
- Brokdorf_Prior3

Constructed priors

- Brokdorf overestimation x2
- Brokdorf overestimation x10
- Brokdorf underestimation x0,5
- Brokdorf underestimation x0,1
- Brokdorf_Prior1 x0,8
- Brokdorf time-dependent source term error (x2, x0,5)
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- Brokdorf local weather turbulences
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- Brokdorf area-wise strong random noise
- Brokdorf perfect
- Emsland instead of Brokdorf (completely wrong)

Original scenarios and prior scenarios from the RODOS system

As part of this research project, two RODOS simulations are used as original scenarios, which are to be reconstructed using the final mechanism of the method Cuboid Splatting with prior: the Emsland scenario and the Brokdorf scenario. These two scenarios, which originate from the RODOS system, are assumed to be the ground truth of the course of the radiological emergency.

There are additional prior scenarios from the RODOS system for each of these original scenarios. These are the EuRim-Emsland scenario as a prior for the Emsland scenario and the Brokdorf_Prior1, Brokdorf_Prior2, and Brokdorf_Prior3 scenarios as priors for the Brokdorf scenario. When generating these RODOS scenarios, the input parameters were varied in order to model a prior scenario in the form of a RODOS scenario that already contains useful information but also errors, which is used as the starting point for a reconstruction.

The Emsland scenario was used in a previous research project by Friedrich et al. (2026) as well, while the Brokdorf scenario and the associated prior scenarios were generated by the domain experts for radiation protection specifically for this research project. The two original scenarios and their associated prior scenarios are presented in more detail below.

The Emsland original scenario

The Emsland scenario was calculated for the location of the former Emsland nuclear power plant and a fictitious accident beginning at 12 noon on March 22, 2023, using the RODOS system. The source term “FKF” from the source term library of the Federal Office for Radiation Protection (Löffler, et al. 2012) was used for this calculation. This source term library contains modeled characteristic release patterns for various types of accidents in nuclear power plants. This scenario covers 116 hours, with only a small amount of radioactive iodine isotopes being released during the first 45 hours, followed by the main release. The weather conditions were modeled using ICON-D2 weather data from the German Weather Service (Reinert, et al. 2023), and the weather situation was fairly stable over the period, with a prevailing wind direction toward the northeast. The Lagrangian particle dispersion model LASAT was used for the dispersion estimate. The area covered by the Emsland scenario is of the size 384 km * 384 km. A spatial resolution of 500 m was used for the entire model domain. The temporal resolution is 1 hour. The iodine concentrations in the air in the Emsland scenario are significantly lower than in the Brokdorf scenario presented below.

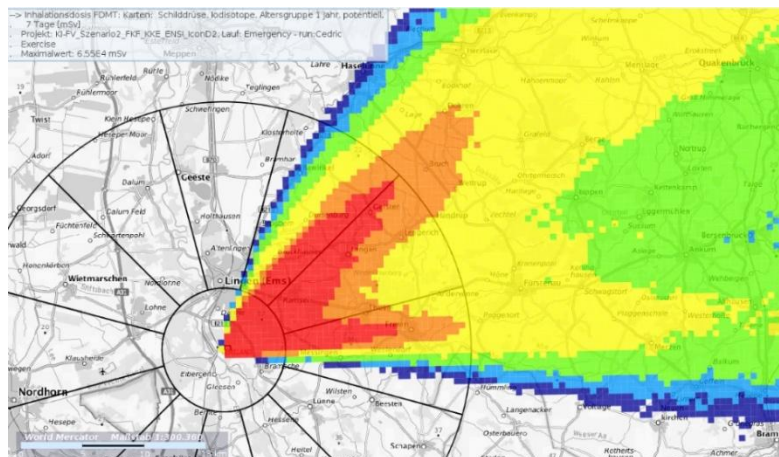


Figure 1: Original scenario Emsland – Preview image from the RODOS system

The prior scenario EuRim-Emsland

The RODOS simulation for the EuRim-Emsland Prior scenario used the same source term (“FKF”) and release period as in the original scenario. The difference in modeling lies, on the one hand, in the weather data used. For this prior scenario, the ICON-EU weather data from the German Weather Service (Reinert, et al. 2023) was used. Secondly, a different type of dispersion model was used. RIMPUFF is a Gaussian puff model and, due to the underlying physical processes, provides less accurate results than a Lagrange particle model. In addition, the spatial resolution

of the prior scenario becomes coarser at greater distances from the release site (“nesting”), so that a resolution of 500 m is only achieved within the innermost 5 km. Overall, this prior scenario has a scenario error of approximately 79% in the resolutions (10x10, 30x30) and 92% in the resolution 100x100 compared to the original scenario.

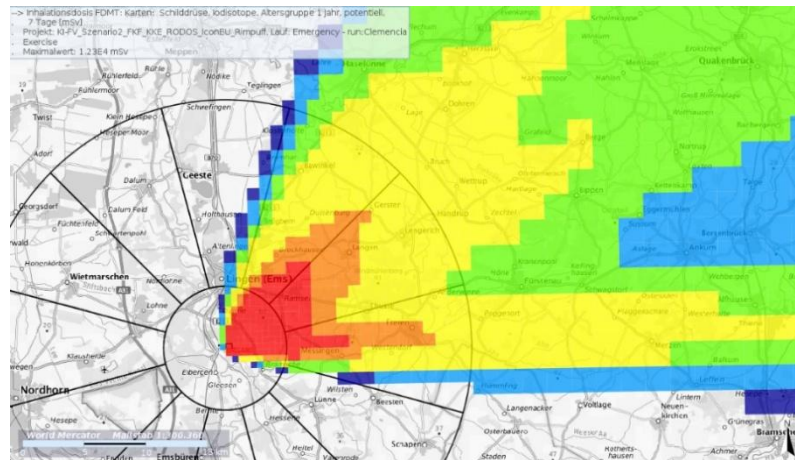


Figure 2: Prior scenario EuRim-Emsland – Preview image from the RODOS system

The Brokdorf original scenario

The Brokdorf scenario was calculated for the former Brokdorf nuclear power plant site and a fictitious accident beginning at 7 p.m. on March 17, 2025, using the RODOS system. The source term “FKA” from the source term library of the Federal Office for Radiation Protection (Löffler, et al. 2012) was used. This scenario spans 60 hours, with the main release of radioactive iodine isotopes beginning directly in the first few hours. Weather conditions were modeled using ICON-D2 weather data from the German Weather Service (Reinert, et al. 2023). Compared to the Emsland scenario, weather conditions are less stable in this scenario, as the prevailing wind direction fluctuates more. The Langrangian particle dispersion model LASAT was used for the dispersion estimate. The area covered by the Brokdorf scenario is of the size 768 km * 768 km. A spatial resolution of 250 m was used for the entire model domain. The temporal resolution is 1 hour. The iodine concentrations in the air in the Brokdorf scenario are significantly higher than in the Emsland scenario presented above, because the FKA source term describes a more severe emergency with a higher total release of radioactivity than the FKF source term.

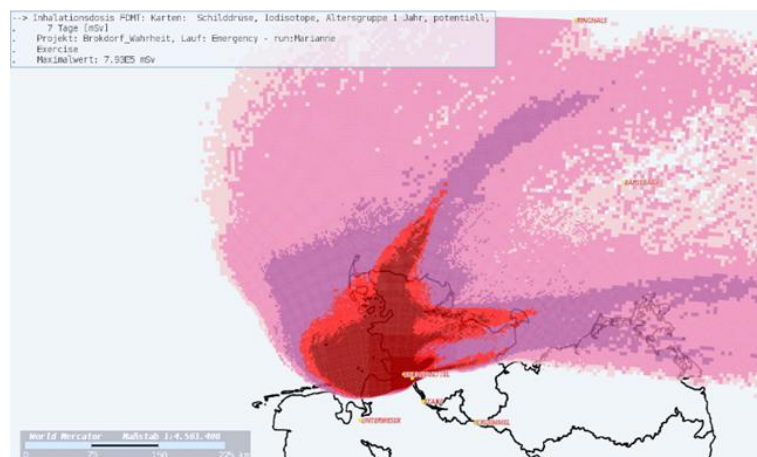


Figure 2: Original scenario Brokdorf – Preview image from the RODOS system

The prior scenario Brokdorf_Prior1

The RODOS calculation for the Brokdorf_Prior1 prior scenario used the same source term (“FKA”), release period, and dispersion model (LASAT). The difference in modeling lies in the weather data used. The ICON-EU weather data from the German Weather Service (Reinert, et al. 2023) was used for this prior scenario. The numerical weather prediction model ICON can be operated at different resolutions: ICON-EU weather data used for this run has a coarser spatial resolution (approx. 6.5 km) than ICON-D2 weather data (approx. 2.2 km) and therefore provides less accurate results (for Germany). There are therefore slight deviations in the weather conditions taken into account in the dispersion estimate. In addition, the prior scenario is more coarsely gridded outward from the location of the nuclear power plant. Overall, this prior scenario has a scenario error of approximately 19% in the 10x10 resolution, 25% in the 30x30 resolution, and 43% in the 100x100 resolution compared to the original scenario.

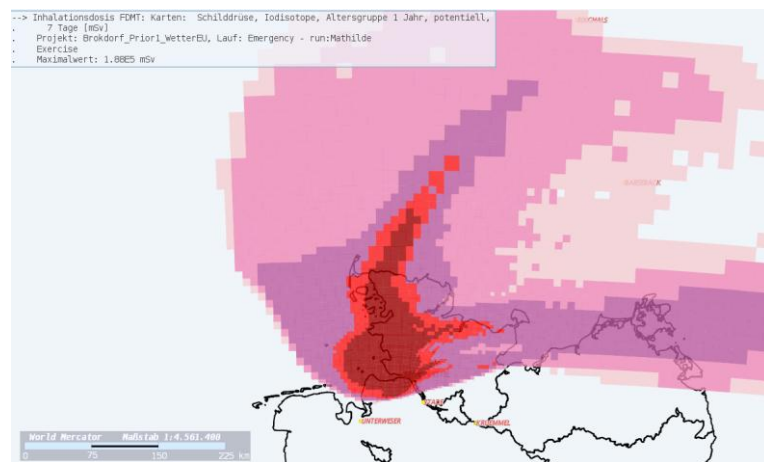


Figure 3: Prior scenario Brokdorf_Prior1 – Preview image from the RODOS system

The prior scenario Brokdorf_Prior2

The RODOS calculation for the Brokdorf_Prior2 prior scenario used the same source term (“FKA”), the same weather data (ICON-D2), and the same dispersion model (LASAT) as for the original Brokdorf scenario. The difference lies in the release period. Here, the start of the release was modeled as being brought forward by five hours (i.e., 2 p.m. instead of 7 p.m.). There is therefore a systematic error along the time axis and thus in relation to the temporal course of the dispersion. In addition, this prior scenario also has a coarser resolution on the outside. Overall, the prior scenario thus has a scenario error of approximately 67% in the 10x10 resolution, 72% in the 30x30 resolution, and 78% in the 100x100 resolution compared to the original scenario.

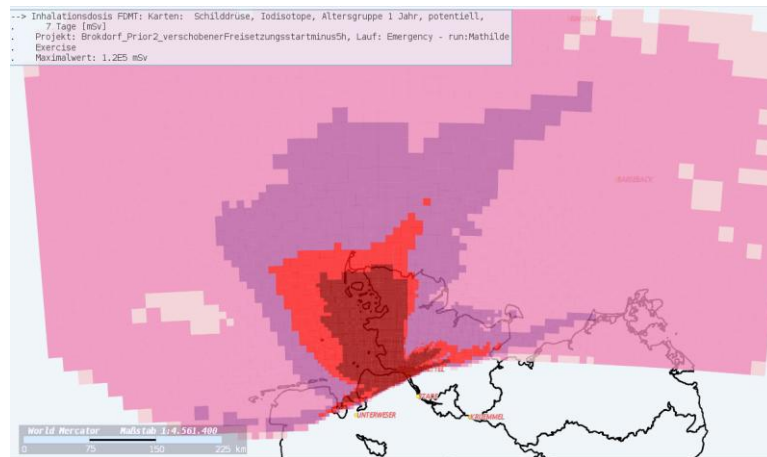


Figure 4: Prior scenario Brokdorf_Prior2 – Preview image from the RODOS system

The prior scenario Brokdorf_Prior3

The RODOS calculation for the Brokdorf_Prior3 prior scenario used the same source term (“FKA”), release period, and weather data (ICON-D2) as for the original Brokdorf scenario. The difference in the modeling lies in the dispersion model used. The older RIMPUFF model (a Gaussian puff model) was used instead of the current LASAT model (a Langrange model). This means that there are deviations in the assumed dispersion of iodine isotopes in the air. In addition, this prior scenario is also more coarsely gridded towards the outside. Overall, the prior scenario thus has a scenario error of approximately 34% in the 10x10 resolution, 41% in the 30x30 resolution, and 67% in the 100x100 resolution compared to the original scenario.

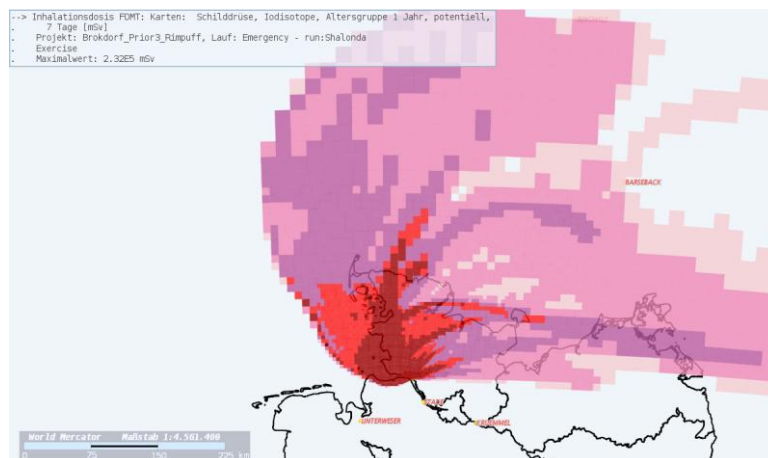


Figure 5: Prior scenario Brokdorf_Prior3 – Preview image from the RODOS system

Targeted construction of prior scenarios

For the targeted construction of prior scenarios containing various types of errors as presented in the previous chapter, a series of functions was implemented, which can be found in the Code Supplement in the directory prior_construction. The modeling of the individual error types is presented below.

The construction of the priors is always based on the original scenarios and prior scenarios presented above. The original scenarios and prior scenarios already gridded to specific resolutions of space-time were used. Priors were generated for all three spatial resolutions, 10x10, 30x30, and 100x100.

Source term errors – Systematic over- or underestimation of the release quantity

One type of constructed prior involves a systematic overestimation or underestimation of the release quantities in the source term. For this purpose, the function `multiply_scenario_by_factor()` was implemented, which multiplies all iodine concentrations contained in a scenario by a specific factor.

For the priors constructed in this way, based on the two original scenarios Emsland and Brokdorf, an overestimation and underestimation of the same magnitude (doubling and halving) and one of a different magnitude (tenfold and one-tenth) of the iodine concentrations were generated.

In addition, this type of error was also applied to the EuRim-Emsland prior and the Brokdorf_Prior1 prior by multiplying them by a factor of 0.8. This was done in order to introduce an additional quantity error into the source term in these priors, as they contain the same source term as the respective original scenario.

This resulted in the following constructed priors for the evaluation:

Emsland:

- Emsland overestimation x2
- Emsland overestimation x10
- Emsland underestimation x0,5
- Emsland underestimation x0,1
- EuRim prior x0,8

Brokdorf:

- Brokdorf overestimation x2
- Brokdorf overestimation x10
- Brokdorf underestimation x0,5
- Brokdorf underestimation x0,1
- Brokdorf_Prior1 x0,8

Source term errors – Time dependent over- and underestimation of the release quantity

The next type of constructed priors also relates to errors in the source term, more specifically to the estimation of the release quantity or the release pattern. In individual time steps of a scenario, quantity overestimations or underestimations are constructed, for which the function `multiply_scenario_at_some_timesteps()` was implemented. In two sets of three consecutive time steps of a scenario, all iodine concentrations are overestimated by a certain factor, and in two sets of three consecutive time steps, all iodine concentrations are underestimated by a certain factor.

The priors were constructed based on the original Emsland and Brokdorf scenarios, using factors for overestimation and underestimation in the same order of magnitude (2 and 0.5) on the one hand, and in a different order of magnitude (10 and 0.1) on the other.

This resulted in the following constructed priors for the evaluation:

Emsland:

- Emsland time-dependent source term error (x2, x0,5)
- Emsland time-dependent source term error (x10, x0,1)

Brokdorf:

- Brokdorf time-dependent source term error (x2, x0,5)
- Brokdorf time-dependent source term error (x10, x0,1)

Weather data errors – modeling errors caused by local turbulence

The function `create_local_weather_turbulences()` was implemented to model errors in weather data caused by small-scale turbulence. Local turbulence is introduced at randomly selected locations in the scenario by rotating 2x2 space-time cells by 90 degrees, 180 degrees, or 270 degrees along the spatial dimensions (x, y). Since the original scenarios rasterized into different

resolutions are used, the number of locations at which such local rotations of iodine concentration values are performed was increased with the resolution in order to achieve a similar error level in terms of the scenario error in the different resolutions. The NumPy function `np.rot90()` was used to implement the local rotation.

This resulted in the following constructed priors for the evaluation:

- | | |
|---|--|
| Emsland: | Brokdorf: |
| <ul style="list-style-type: none"> • Emsland local weather turbulences | <ul style="list-style-type: none"> • Brokdorf local weather turbulences |

Weather data errors – Modeling a systematic error in wind speed in the main wind direction

To model a systematic error in the weather data, the systematic overestimation of wind speed was constructed in a main wind direction of the respective original scenario. To do this, the main wind directions were first determined for each scenario by generating a heat map of the iodine concentrations summed over the time axis. In both cases, the upper right quadrant was identified as a rough main wind direction based on the centrally located nuclear power plant. Then, the function `stretch_towards_main_wind_direction()` was implemented, which stretches the iodine concentration values contained in this quadrant from the nuclear power plant in the rough main wind direction. This happens for each time step of the scenario. In addition, each time step is normalized with the sum of the previously contained iodine concentrations of that time step so that no more iodine air pollution is created per time step than before, but is simply distributed differently in the area due to the stretching. For the stretching, the implementation of an affine transformation of the SciPy image transformation library `Ndimage` was used.

This resulted in the following constructed priors for the evaluation:

- | | |
|---|--|
| Emsland: | Brokdorf: |
| <ul style="list-style-type: none"> • Emsland systematic wind speed error | <ul style="list-style-type: none"> • Brokdorf systematic wind speed error |

Prior quality levels – Area-wise random noise added to the prior

To generate different prior qualities based on the respective original scenarios for Emsland and Brokdorf, the scenarios were each transformed using random noise of varying intensity. Instead of applying this noise pointwise, which would mean applying a different noise factor for each space-time cell, the noise was applied to three-dimensional subregions of space-time. This was done so that the noise would not disrupt the entire systematics contained in the scenario from the dispersion model and the weather model. For each area, a random factor from a uniform distribution from a specific interval is applied to the iodine concentrations present in the scenario. For each scenario, a prior with very low random noise was generated, for which the factors were drawn from the interval $[0.95; 1.05]$. In addition, a prior with medium random noise was generated, for which the factors were taken from the interval $[0.5; 1.5]$. There is also a prior with strong random noise, where the factors for the Emsland scenario are taken from the interval $[0.0; 3.5]$ and for the Brokdorf scenario from the interval $[0.0; 7.5]$. The sizes of these intervals were determined experimentally in order to achieve a specific level of scenario errors in the constructed priors. These are $<5\%$ scenario error for low noise, around 20 to 30% scenario error for medium noise, and $>100\%$ scenario error for very high noise.

This resulted in the following constructed priors for the evaluation:

- | | |
|--|---|
| Emsland: | <ul style="list-style-type: none"> • Emsland area-wise medium random noise |
| <ul style="list-style-type: none"> • Emsland area-wise small random noise | |

- Emsland area-wise strong random noise

Brokdorf:

- Brokdorf area-wise small random noise

- Brokdorf area-wise medium random noise

- Brokdorf area-wise strong random noise

Extreme prior qualities – Perfect or completely wrong prior

In order to test the two extremes of prior qualities, a perfect prior and a completely wrong prior were also generated for each original scenario. The original scenario was simply used unchanged as the perfect prior in each case. The two original scenarios were swapped to create completely wrong Priors, so that although the scenario still involved the spread of iodine isotopes from a nuclear power plant located in the center of the scenario, the release quantity, pattern, and weather data were completely wrong. The `replace_emsland_brokdorf()` function was implemented for this purpose. Since both scenarios comprise different numbers of time steps (Emsland 116 hours, Brokdorf 60 hours), the time steps suitable for the exchange were first analyzed by summing the iodine concentrations along the x and y axes for each time step. It was found that the majority of the release in the Emsland scenario takes place between time steps 40 and 100, so these time steps were used in the Brokdorf scenario. Conversely, the Brokdorf scenario was used in approximately this area of the Emsland scenario, so that the most heavily contaminated time steps roughly correspond there as well.

This resulted in the following constructed priors for the evaluation:

Emsland:

- Emsland perfect
- Brokdorf instead of Emsland (completely wrong)

Brokdorf:

- Brokdorf perfect
- Emsland instead of Brokdorf (completely wrong)